

University of Science and
Technology in Zewail City

TOPIC 2

TEMPERATURE (T) VS. CARRIER CONCENTRATIONS AND VBI

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INTRODUCTION

Temperature (T) vs. Carrier Concentrations (n_n , n_p , p_p , p_n , n_i) and V_{bi}

This project investigates the effect of temperature variation on carrier concentrations (n_n , n_p , p_p , p_n , n_i) and the built-in potential (V_{bi}) of an abrupt Silicon p-n junction.

Using simulation, the temperature is swept over a range to analyse changes in carrier distribution and junction characteristics, providing a deeper understanding of semiconductor device behaviour under different thermal conditions.

STRATEGIC GOALS

01

Temperature

- Individual Run: 400 K–
- Sweep: 25 → 900 K

02

Donor Concentration

- Individual Run: 1×10^{18}
/cm³
- Sweep: 1×10^{18} /cm³

03

Acceptor Concentration

- Individual Run: 1×10^{17}
/cm³
- Sweep: 1×10^{17} /cm³

SIMULATION RESULTS

After that we did the sweep over a different temperature range and all results and data are provided in the zip file.

```
ATLAS> EXTRACT> init infile="Topic2LogFile.log"
EXTRACT> extract name="Majority Electron Concentration (/cm^3)" max(curve(v."anode", probe."nn"))
Majority Electron Concentration (/cm^3)=8.72508e+017
EXTRACT> extract name="Majority Hole Concentration (/cm^3)" max(curve(v."anode", probe."pp"))
Majority Hole Concentration (/cm^3)=9.21562e+016
EXTRACT> extract name="Minority Electron Concentration (/cm^3)" max(curve(v."anode", probe."np"))
Minority Electron Concentration (/cm^3)=4.53383e+008
EXTRACT> extract name="Minority Hole Concentration (/cm^3)" max(curve(v."anode", probe."pn"))
Minority Hole Concentration (/cm^3)=1.00291e+008
EXTRACT> extract name="Intrinsic Carrier Concentration (/cm^3)" max(curve(v."anode", sqrt(probe."nn"*probe."pn")))
Intrinsic Carrier Concentration (/cm^3)=9.35438e+012
EXTRACT> extract name="V_bi" max(curve(v."anode", probe."Max Potential"-probe."Min Potential"))
V_bi=0.724301
```

Figure 1: Results of an individual run

RESEARCH OUTPUT AND GRAPH

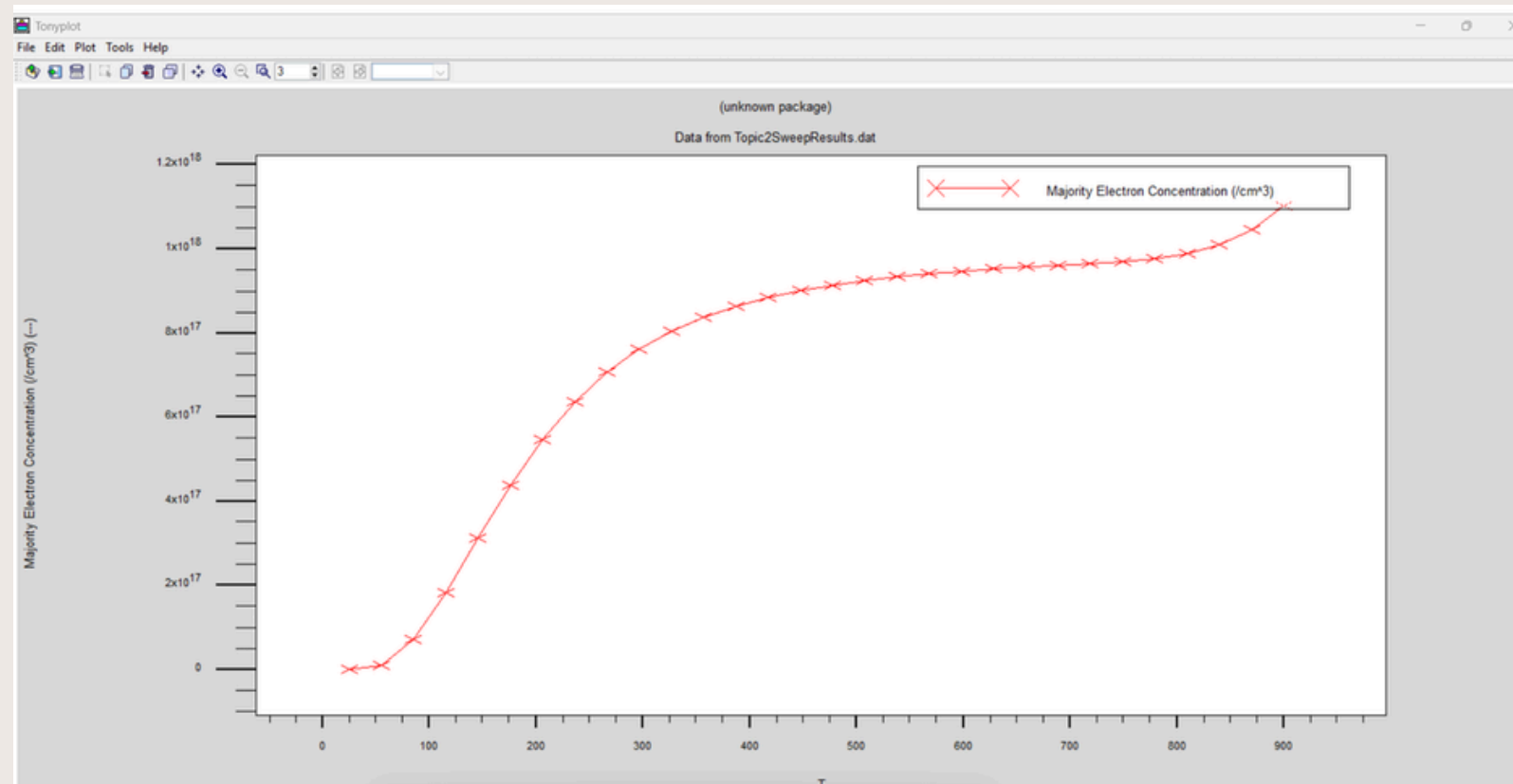


Fig1:
**Majority electron
concentration vs T**

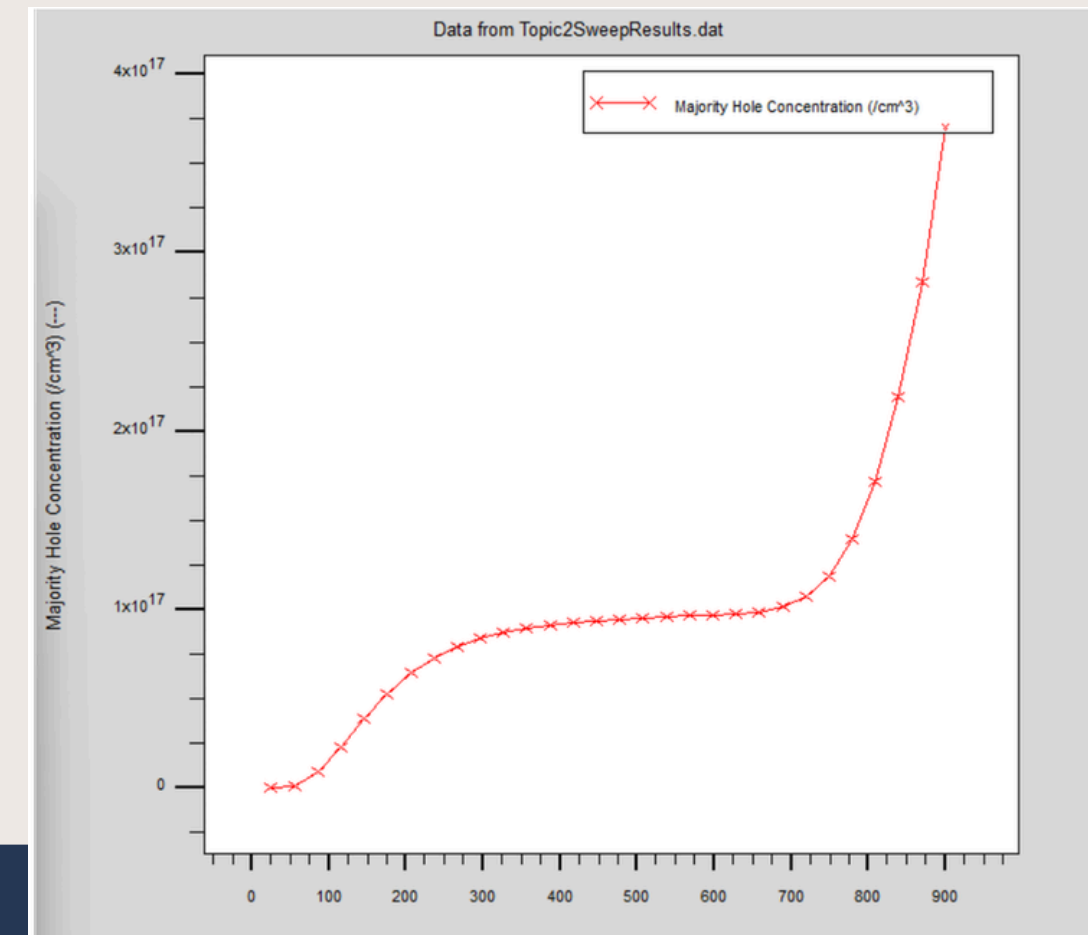


Fig2:
**Majority hole
concentration vs T**

OUTPUTS AND GRAPHS

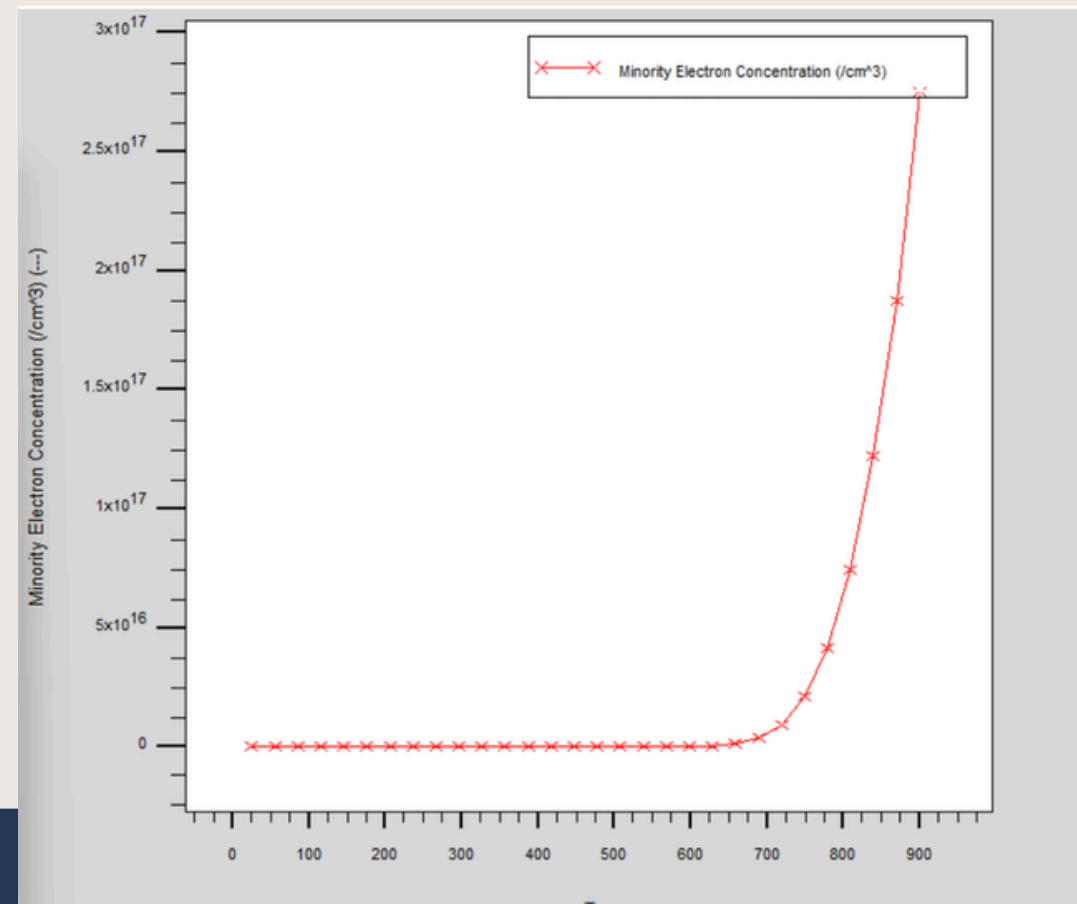


Fig3:
**Minority electron
concentration vs T**

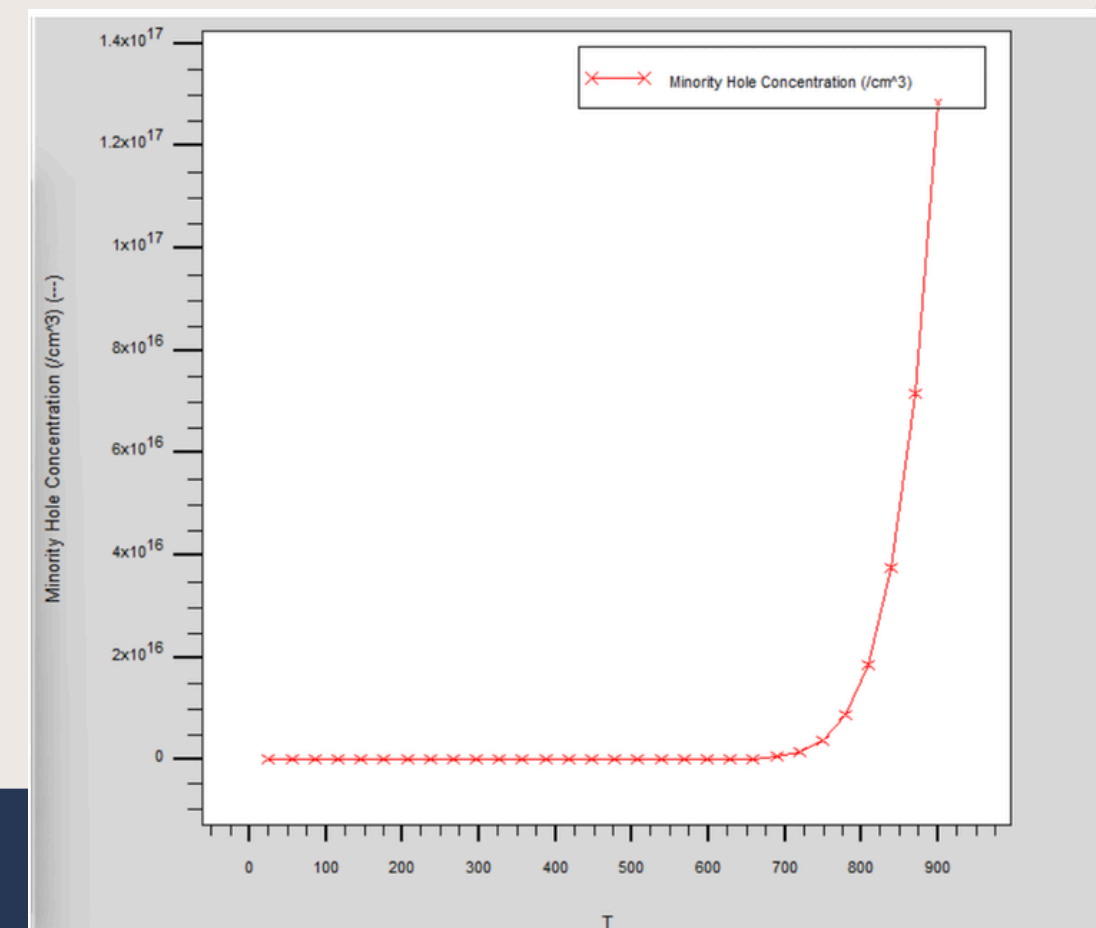


Fig4:
**Minority hole
concentration vs T**

OUTPUTS AND GRAPHS

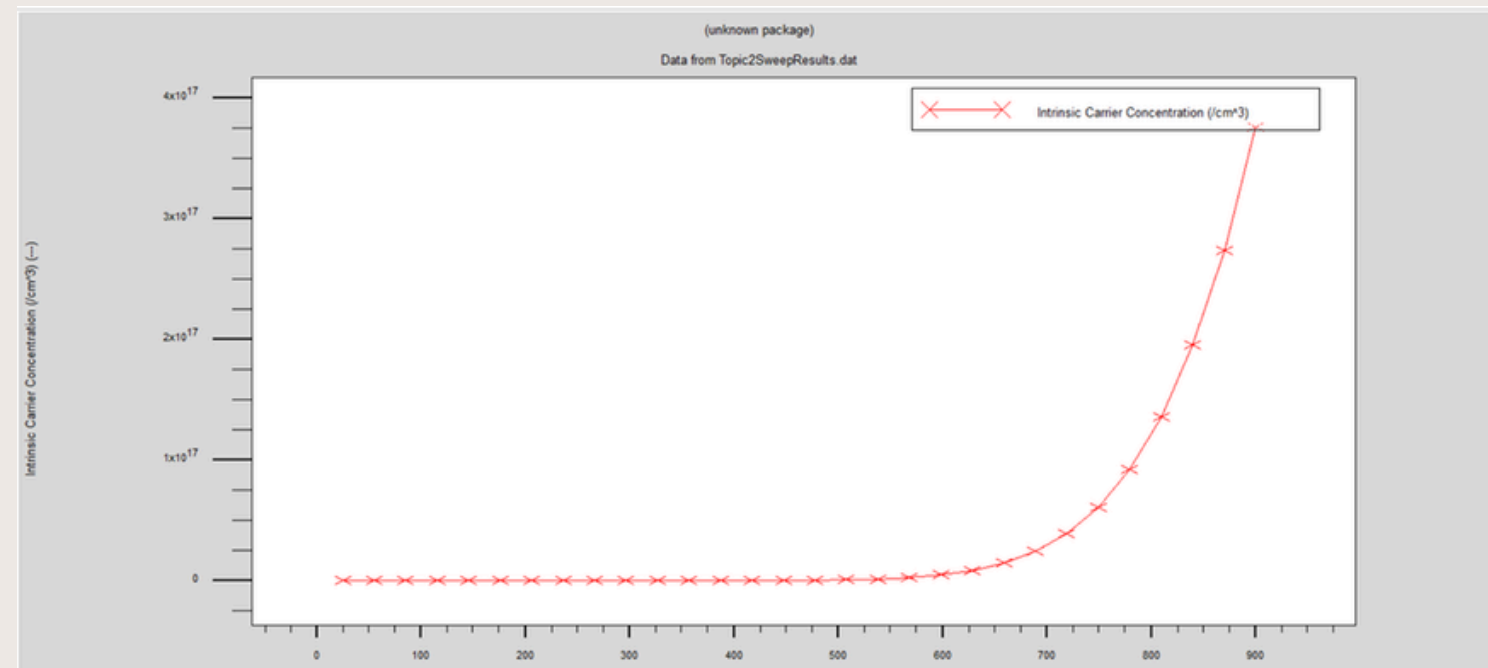


Fig5:
**Intrinsic carrier
concentration vs T**

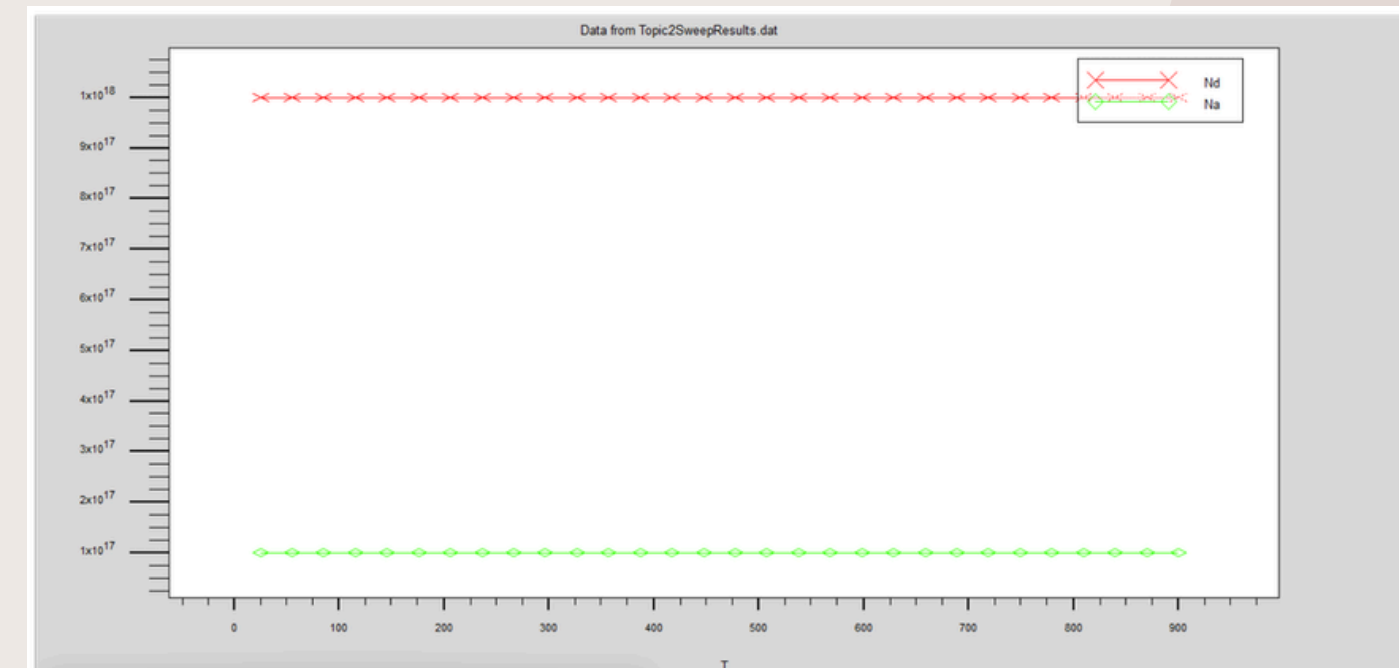
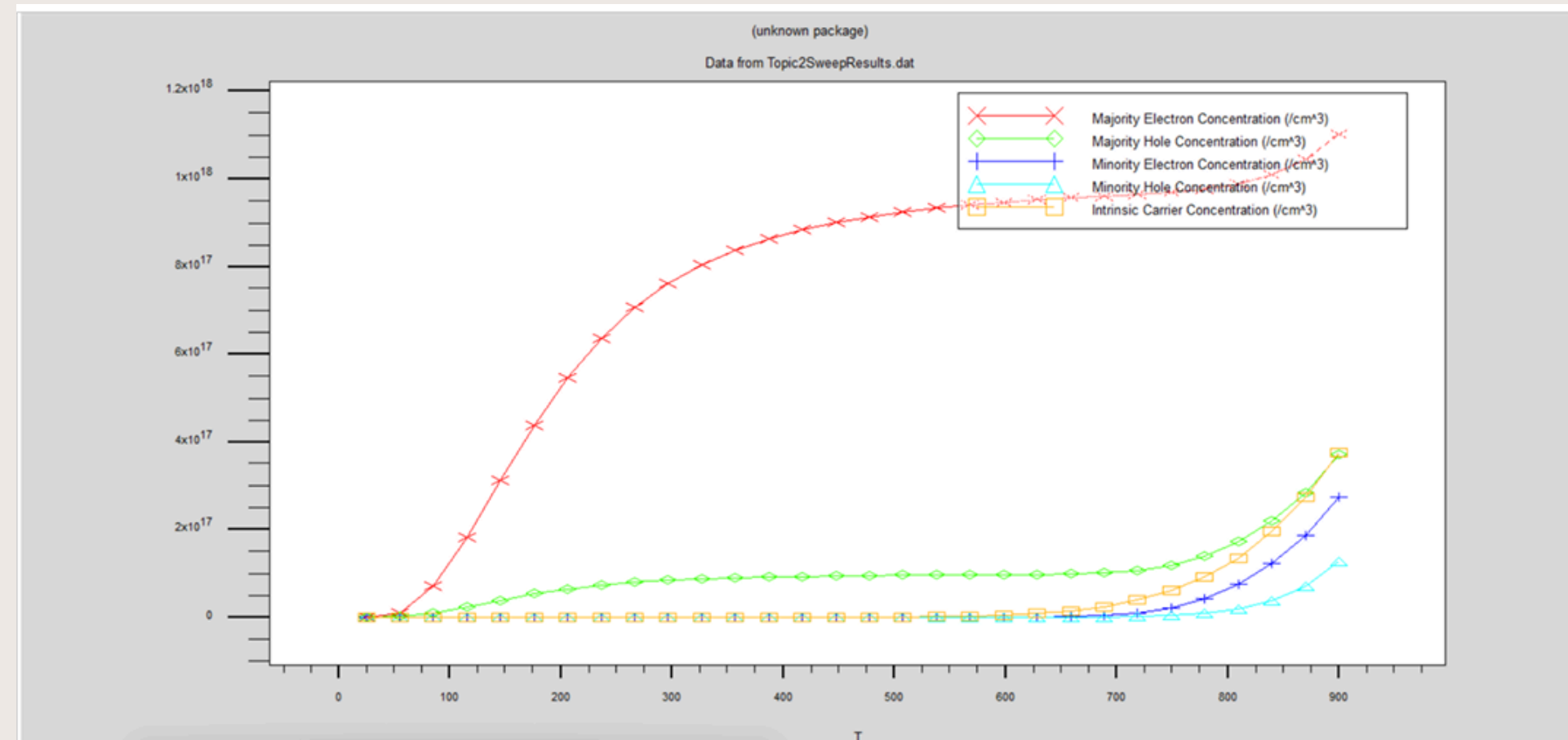


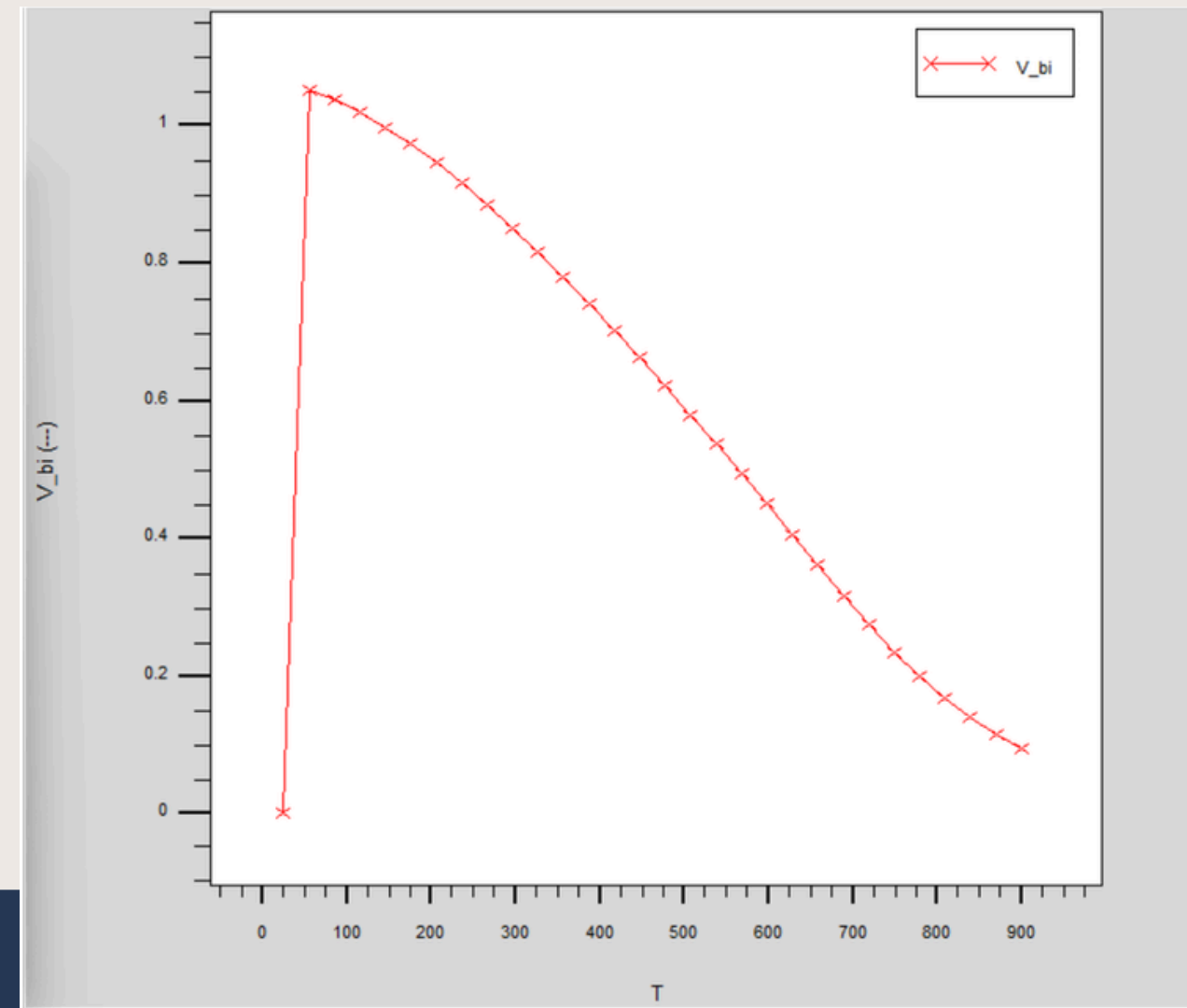
Fig6:
 N_d and N_a vs T

OUTPUTS AND GRAPHS



**Carriers Concentration vs.
Temperature in a Doped
Semiconductor**

OUTPUTS AND GRAPHS



V_{bi} vs temperature
(#first node is very close to 0
 $v_{bi}(0)=0$)

RESEARCH OUTPUT AND EXPLANATIONS

The similarity between these two figures is that the majority electron concentration goes through four phases, and what determines them is the temperature

First phase: freeze out

This happens when $T=0K$, so there are no free carriers to move.

Second phase: partial ionization

Increasing temperature generates more electron-hole pairs, increasing intrinsic carrier concentration and affecting electron concentration under partial ionisation.

$$n_0 = \frac{N_D}{1 + \frac{1}{2} \exp \left(\frac{E_D - E_F}{kT} \right)}$$

Third phase: Extrinsic

At room temperature (300 K), donor and acceptor atoms are almost completely ionised, so the dopant concentration mainly determines the majority carrier concentration

$$n_0 = N_d$$

Forth phase: Intrinsic

In the intrinsic region, high temperature causes thermally generated electron-hole pairs to dominate semiconductor behavior. The intrinsic carrier concentration (n_i) increases rapidly and eventually exceeds the dopant concentration, making carrier concentrations mainly controlled by thermal generation instead of impurity atoms.

THE BEHAVIOR OF VBI WITH TEMPERATURE IS DESCRIBED BY

$$V_{bi} = \frac{kT}{q} \ln \left(\frac{N_a N_d}{n_i^2} \right)$$

At ($T = 0K$), the built-in potential (V_{bi}) is zero because no free carriers are available. Although (V_{bi}) seems to increase linearly with temperature, the intrinsic carrier concentration (n_i) increases exponentially and dominates the behaviour. As a result, the overall built-in potential decreases with increasing temperature.

The physical interpretation is that as temperature increases, the generation of intrinsic carriers causes the p-type and n-type regions to lose their distinct extrinsic characteristics and behave more like intrinsic silicon.

Temperature-Dependent Equations

$$E_g(T) = E_g(0) - \frac{\alpha T^2}{T + \beta}$$

$$N_c(T) = N_c(300) \left(\frac{T}{300} \right)^{3/2}$$

$$N_v(T) = N_v(300) \left(\frac{T}{300} \right)^{3/2}$$

$$n_i = \sqrt{N_c N_v} e^{-E_g/(2kT)}$$

At $T = 900K$

$$E_g(900) \approx 0.92 \text{ eV}$$

$$N_c(900) \approx 1.46 \times 10^{20} \text{ cm}^{-3}$$

$$N_v(900) \approx 5.40 \times 10^{19} \text{ cm}^{-3}$$

$$n_i \approx 2.36 \times 10^{17} \text{ cm}^{-3}$$

$$n_n \approx 9.57 \times 10^{17} \text{ cm}^{-3}$$

$$p_n \approx 5.72 \times 10^{16} \text{ cm}^{-3}$$

$$n_p \approx 9.61 \times 10^{17} \text{ cm}^{-3}$$

$$V_{bi} \approx 0.047V$$

Analytical

$$n_i \approx 2.41 \times 10^{17} \text{ cm}^{-3}$$

Silvaco

$$n_i \approx 3.76 \times 10^{17} \text{ cm}^{-3}$$

At t=400K

For silicon at 400K:

$$E_g(400K)_{pure} = 1.17 - 0.073 = 1.097 \text{ eV}$$

This calculation only considers the thermal effect on the bandgap.

Silvaco predicts:

$E_g(400K) \approx 1.05 \text{ eV}$ because the simulation includes:

- temperature-induced narrowing,
- heavy-doping bandgap narrowing (BGN).

The ATLAS command:

BGN.KLAASSEN

activates Klaassen's Bandgap Narrowing model.

$$E_g = E_{g0} - \Delta E_g(T) - \Delta E_g(Doping)$$

$$E_g = 1.17 - 0.073 - 0.047$$

$$E_g \approx 1.05 \text{ eV}$$

- Increasing temperature reduces the silicon bandgap.
- Lower E_g increases thermal carrier generation.
- n_{in_ini} rises dramatically at high temperature.
- Minority carrier concentrations increase significantly.
- Built-in voltage decreases with temperature.

At T=400k silvaco

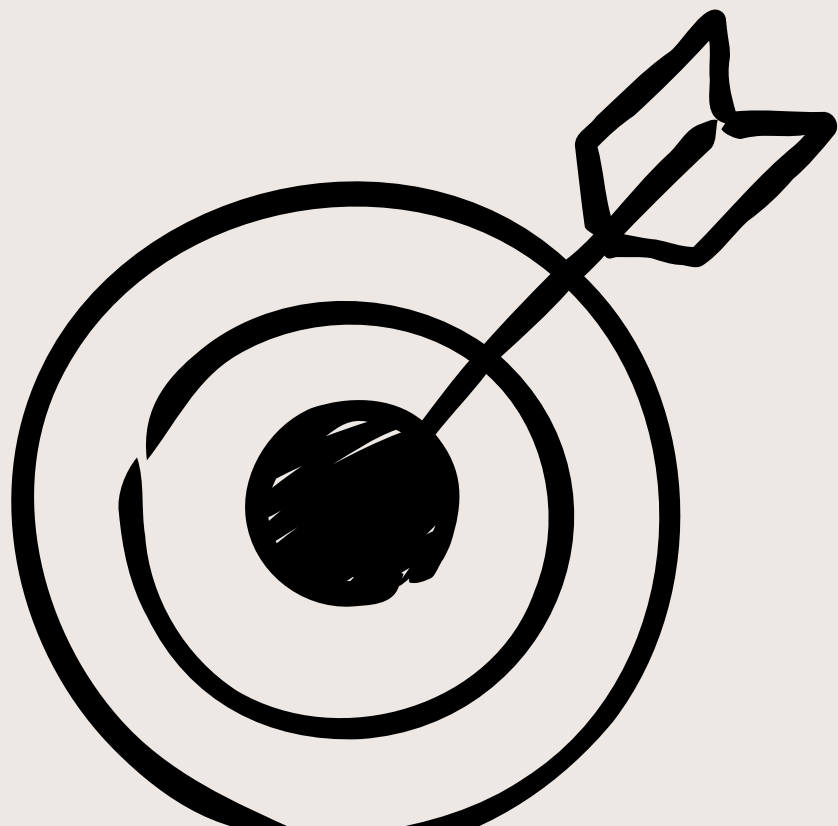
Band Parameters

Epsilon	:	11.8
Eg (eV)	:	1.05
Chi (eV)	:	4.18
Nc (per cc)	:	4.31e+019
Nv (per cc)	:	1.6e+019
ni (per cc)	:	6.16e+012

OUR TEAM

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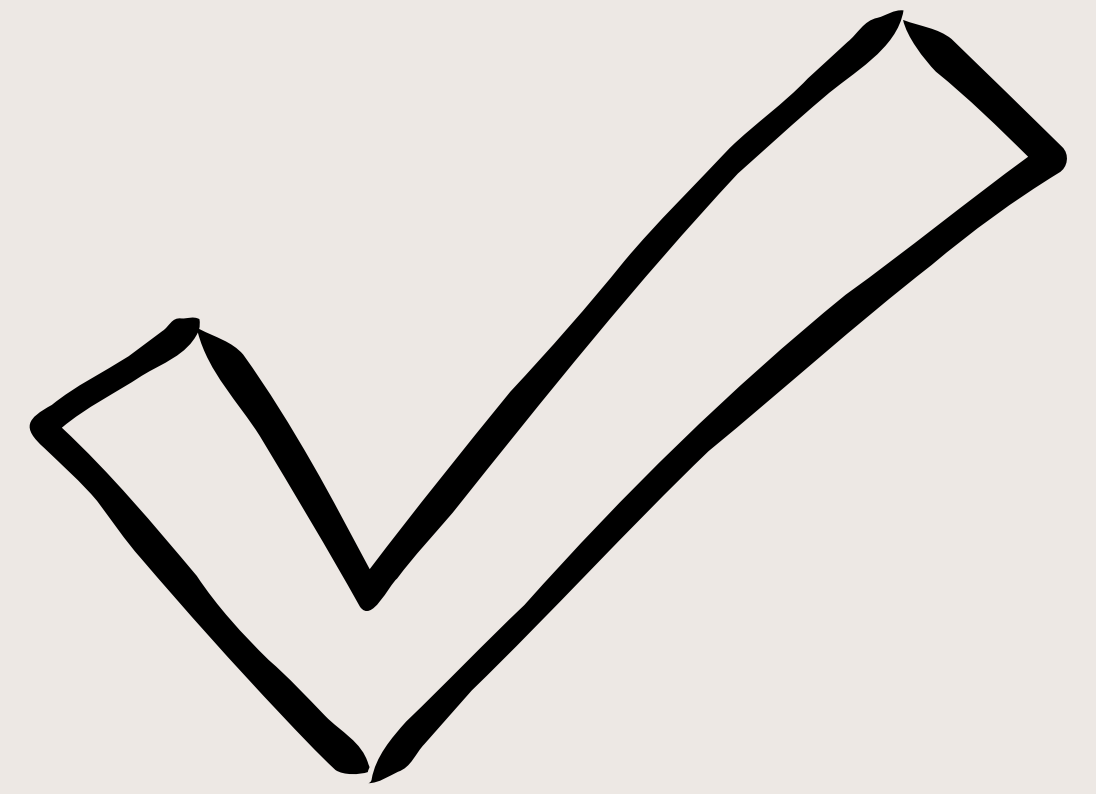
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THANK YOU